Yatin Dandi

Learning and Computation in High Dimensions

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Website · Scholar · GitHub
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FEATURE LEARNING · DEPTH & HIERARCHY · TRAINING DYNAMICS · SCALING LAWS · SPIN GLASSES

Summary

My research asks which structures make high-dimensional problems statistically or algorithmically hard, and which permit efficient learning or computation. In particular, I develop predictive theories of feature learning and the role of depth in neural networks, together with theories of dynamics, optimization, and statistical-to-computational gaps in spin glasses and random graphs. This work draws on mathematical tools from probability, optimization, learning theory, and statistical physics.

Education

2022 – 2027 **EPFL**
PhD in Computer Science · expected August 2027
Advisors: Lenka Zdeborová and Florent Krzakala.

2017 – 2022 **Indian Institute of Technology Kanpur**
BTech–MTech in Computer Science and Engineering
CPI: 9.6/10. Master's CPI: 10/10. Thesis: *Tackling Heterogeneity in Collaborative Machine Learning*, co-supervised by Martin Jaggi and Piyush Rai.

Fellowships and Awards

2026	EPFL AI Center PhD Fellowship	Support for high-impact AI research
2026	Jane Street Graduate Research Fellowship	Honorable Mention for doctoral research
2024	EPFL Doc.Mobility Grant	Fellowship for a research visit at MIT
2022	EDIC Fellowship, EPFL	Support for the first year of doctoral studies
2018, 2019	Academic Excellence Award, IIT Kanpur	Awarded twice for academic performance
2017	Aditya Birla Scholarship	One of 15 engineering scholars from the IITs and BITS Pilani
2017	All India Rank 135, JEE Advanced	National entrance examination, 172,024 candidates

Research Highlights

FEATURE LEARNING

Asymptotics of feature learning in two-layer networks
Developed a new route to universality and exact high-dimensional asymptotics by conditioning on gradient-induced spikes, yielding precise characterizations of generalization and spectrum of learned features. Established how batch size and successive gradient steps govern the acquisition of target directions, and showed that reusing batches makes targets learnable that online-SGD analyses classify as hard. These works motivated a refined classification through symmetries and label transformations, and uncovered novel statistical-to-computational gaps.
ICML 2024 Spotlight · AISTATS 2025 Oral · JMLR 2024 · ICML 2024 · AISTATS 2025

DEPTH & HIERARCHY

Depth as sequential dimension reduction
Led work identifying a computational advantage of depth: each layer filters its input into a richer feature space of lower effective dimension, making new features easier to discover. Follow-ups cover hierarchical and compositional targets, scaling laws, and low-degree spectral filtering.
NeurIPS 2025 Spotlight · ICML 2026 · COLT 2026 · HiLD @ ICML 2026

SPIN GLASSES

Statics, dynamics, and statistical-to-computational gaps in spin glasses
Proved exact asymptotic equations governing Glauber dynamics in the SK model and developed the dynamical cavity method into a rigorous technique for first-order algorithms. Characterized stable local optima in random graphs and spin glasses, probing statistical-to-computational gaps.
COLT 2026 · arXiv 2025 · SLMath highlight

Selected Publications

Full list of publications: Google Scholar

PREPRINT 2026

Neural LoFi with Backward Coupling: A Spectral Theory of Cross-Layer Feature Learning arXiv
Yatin Dandi, Matteo Vilucchio, Hugo Tabanelli, Florent Krzakala

NEURIPS 2026

Deep Learning as Neural Low-Degree Filtering: A Spectral Theory of Hierarchical Feature Learning arXiv · code
Yatin Dandi, Matteo Vilucchio, Luca Arnaboldi, Hugo Tabanelli, Florent Krzakala

COLT 2026

Rigorous Asymptotics for First-Order Algorithms Through the Dynamical Cavity Method arXiv
Yatin Dandi, David Gamarnik, Francisco Pernice, Lenka Zdeborová

Selected Publications, continued

NEURIPS
2025 · Spotlight

The Computational Advantage of Depth: Learning High-Dimensional Hierarchical Functions with Gradient Descent arXiv
Yatin Dandi, Luca Pesce, Lenka Zdeborová, Florent Krzakala

PREPRINT
2025 · Under review

Sequential Dynamics in Ising Spin Glasses arXiv · SLMath
Yatin Dandi, David Gamarnik, Francisco Pernice, Lenka Zdeborová

PREPRINT
2025 · Under review

Asymptotics of Non-Convex Generalized Linear Models in High-Dimensions: A Proof of the Replica Formula arXiv
Matteo Vilucchio, **Yatin Dandi**, Matéo Pirio Rossignol, Cedric Gerbelot, Florent Krzakala

AISTATS
2025 · Oral

A Random Matrix Theory Perspective on the Spectrum of Learned Features and Asymptotic Generalization arXiv
Yatin Dandi, Luca Pesce, Hugo Cui, Florent Krzakala, Yue M. Lu, Bruno Loureiro

AISTATS
2025

Fundamental Computational Limits of Weak Learnability in High-Dimensional Multi-Index Models paper
Emanuele Troiani, **Yatin Dandi**, Leonardo Defilippis, Lenka Zdeborová, Bruno Loureiro, Florent Krzakala

ICML
2024

The Benefits of Reusing Batches for Gradient Descent in Two-Layer Networks arXiv
Yatin Dandi, Emanuele Troiani, Luca Arnaboldi, Luca Pesce, Lenka Zdeborová, Florent Krzakala

ICML
2024 · Spotlight

Asymptotics of Feature Learning in Two-Layer Networks after One Gradient Step arXiv
Hugo Cui, Luca Pesce, **Yatin Dandi**, Florent Krzakala, Yue M. Lu, Lenka Zdeborová, Bruno Loureiro

JMLR
2024

How Two-Layer Neural Networks Learn, One (Giant) Step at a Time paper · arXiv
Yatin Dandi, Florent Krzakala, Bruno Loureiro, Luca Pesce, Ludovic Stephan

PREPRINT
2023 · Under review

Maximally-Stable Local Optima in Random Graphs and Spin Glasses: Phase Transitions and Universality arXiv
Yatin Dandi, David Gamarnik, Lenka Zdeborová

Research Experience

SINCE 2022

PhD Researcher — LAUSANNE, SWITZERLAND

EPFL · Advisors: Lenka Zdeborová and Florent Krzakala

– Developed predictive theories of feature learning in shallow and deep networks using high-dimensional probability, random matrix theory, and statistical physics.
– Contribute more broadly to the mathematics of spin glasses and diffusion models, statistical-to-computational gaps, high-dimensional statistics, and random graphs.

2024 – 2025

Visiting Researcher / Program Associate — CAMBRIDGE AND BERKELEY, USA

MIT and Simons Laufer Mathematical Sciences Institute · Host: David Gamarnik

– Undertook an EPFL Doc.Mobility-funded visit at MIT and participated in SLMath's program on Probability and Statistics of Discrete Structures.
– Developed novel techniques for analyzing first-order optimization algorithms in high dimensions and derived exact dynamical equations for Glauber dynamics in the SK model.

SUMMER 2022

Research Intern — MONTREAL, CANADA

Mila · Advisors: Simon Lacoste-Julien and Yoshua Bengio

– Developed a theoretical framework for analyzing convergence of min-max algorithms, including the saddle-point Frank-Wolfe method.

2019 – 2022

Undergraduate Researcher — INDIA AND SWITZERLAND

IIT Kanpur and EPFL · Collaborators: Martin Jaggi, Piyush Rai, Abhishek Kumar

– Developed optimization methods for heterogeneous distributed learning, yielding work on implicit gradient alignment (AAAI 2022) and decentralized mixing (NeurIPS 2022 workshop).
– Developed generalized adversarially learned inference (AAAI 2021) and co-developed a joint image-video generation model (WACV 2020).

Talks and Service

SELECTED TALKS

HiLD Workshop, Banff, 2026 · NYU Center for Data Science, 2026 · Youth in High-Dimensions, Trieste, 2026 · DACO Seminar, ETH Zurich, 2026 · ProbAI Scaling Laws, Warwick, 2026 · Physics of ML, Stanford, 2025 · SLMath Seminar, Berkeley, 2025 · JMM, Seattle, 2025 · Oberwolfach, 2023

SERVICE

Reviewer for Annals of Statistics, NeurIPS, ICML, ICLR, AISTATS, TMLR, and SIAM Journal on Mathematics of Data Science. Supervised several master's-level research projects at EPFL. Teaching assistant for Statistical Physics III, Statistical Physics for Optimization and Learning, Statistical Physics of Computation, and Elements of Statistics for Data Science at EPFL, and for Mathematics for Computer Science II/III and Introduction to Machine Learning at IIT Kanpur.